

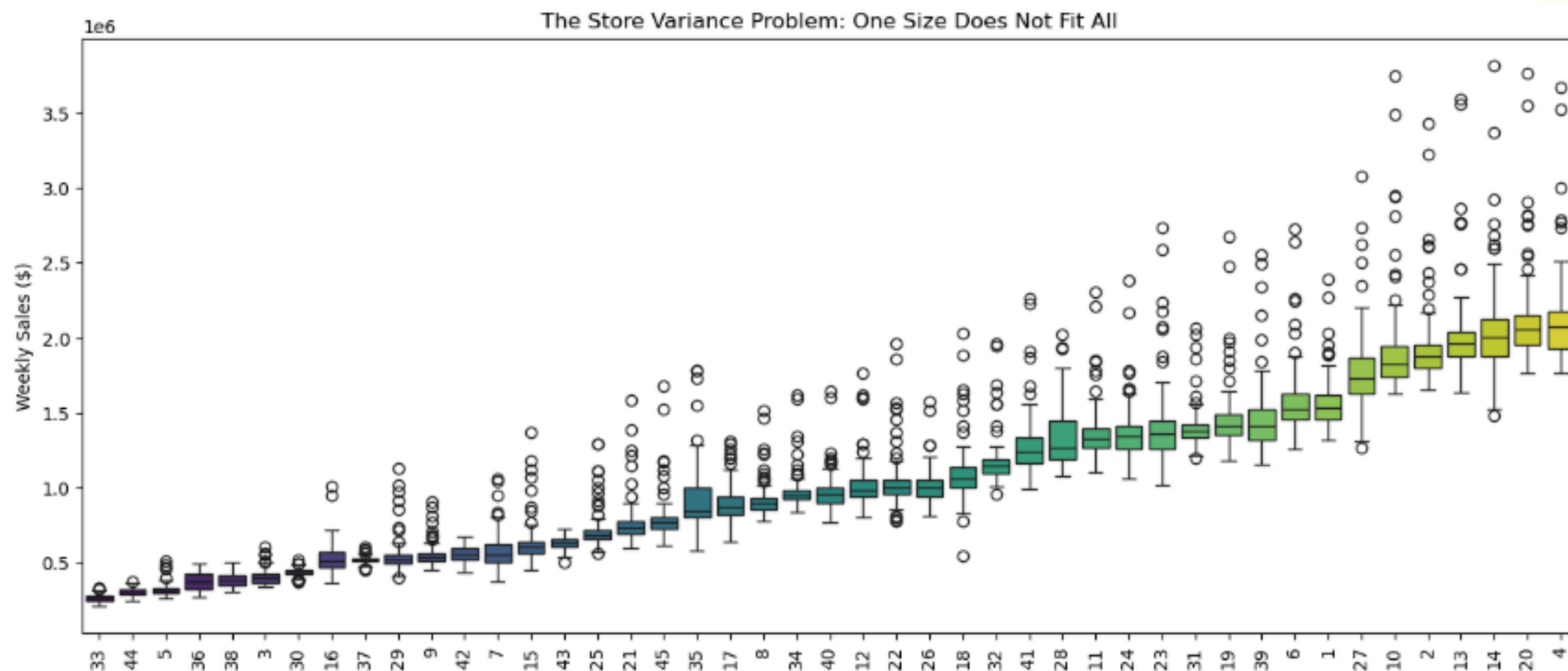
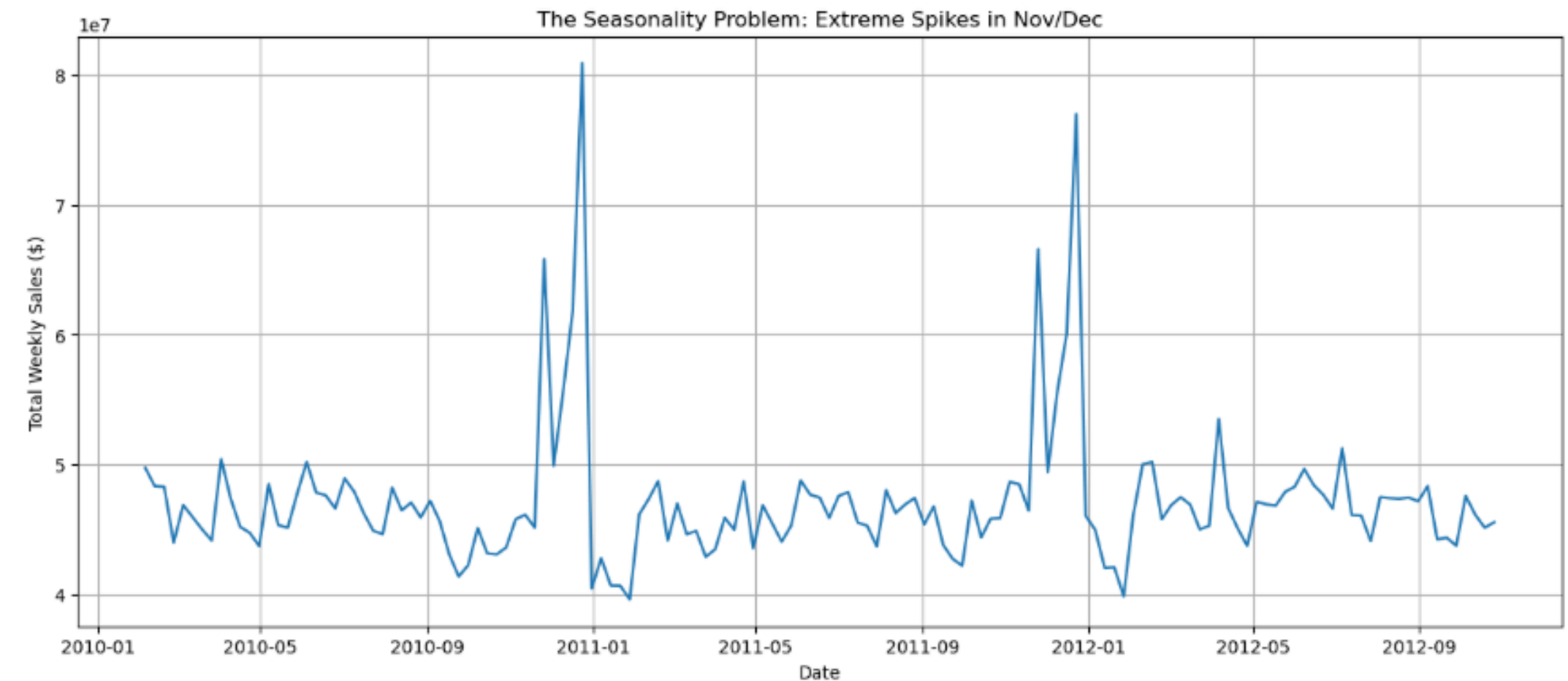
MAXIMISING PROFIT AT Walmart

Forecasting sales using Linear Regression : A data
driven approach to Inventory and staffing

ROSHAN PRAKASH



The Business Problem: One size doesn't fit all



Walmart faced an overstocking issue in 2022. Had to give upto 70% discounts for stock clearance of 1.5 billion \$.

- Retail margins often razor thin.
- Overstocking → Capital wastage (Walmart also sells a lot of fast expiring goods)
- Understocking: Losing revenue to competitors like Target
- Many features which may affect sales, like num of holidays in the week, temperature, fuel price etc..

Very Important to predict weekly sales for precise inventory planning

Data Discovery

Walmart Sales Dataset



SCAN FOR ACCESSING DATA

6435 WEEKS

45 STORES

Target variable (Y) : Weekly sales

Store Attributes(Store ID (dummy),
Holiday or not)

Macro Economic Features(Fuel Price,
Temperature)

Time Features (Month (dummy), Day_status)

90% → Train | 10% → Test

ALGORITHM : OLS Linear Regression

**Validation: R^2 , MAPE, Linearity,
Homoscedasticity, Multicollinearity**

EVOLUTION OF THE MODEL

1

Initial Model

R^2 value: 0.93

MAPE (Train): 9.12%

MAPE (Test): 9.86%

2

Log model

R^2 value: 0.96

MAPE (Train): 7.18%

MAPE (Test): 7.58%

3

Final model

R^2 value : 0.96

MAPE (Train): 7.21%

MAPE (Test): 7.59%

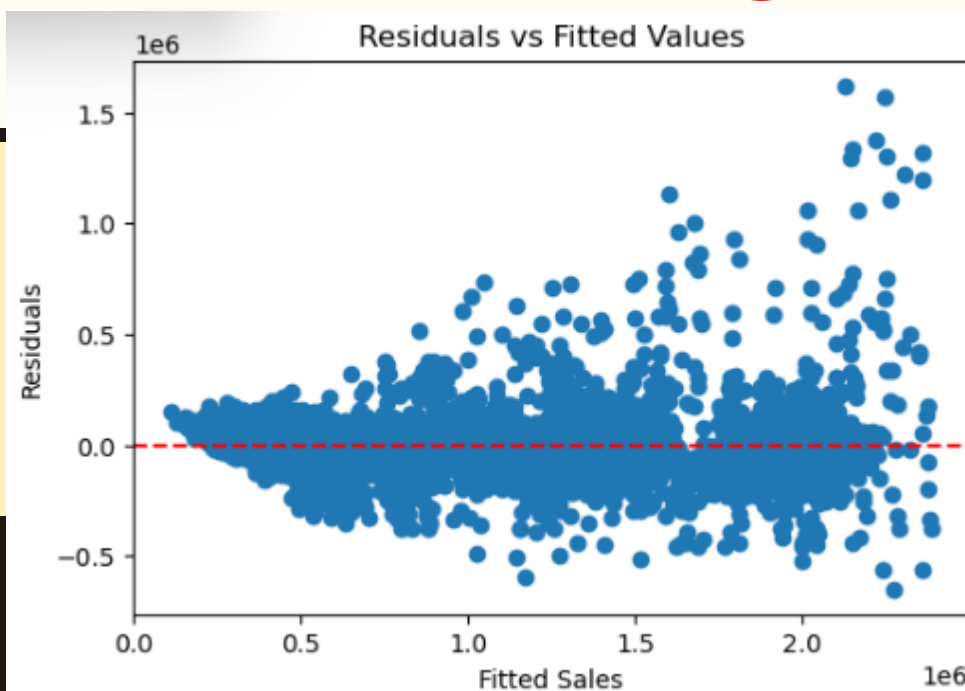
$\xrightarrow{\text{np.log}(Y)}$

$\xrightarrow{\text{Remove CPI and Temperature}}$

Validation

Linear ✓

Homoscedastous ✗

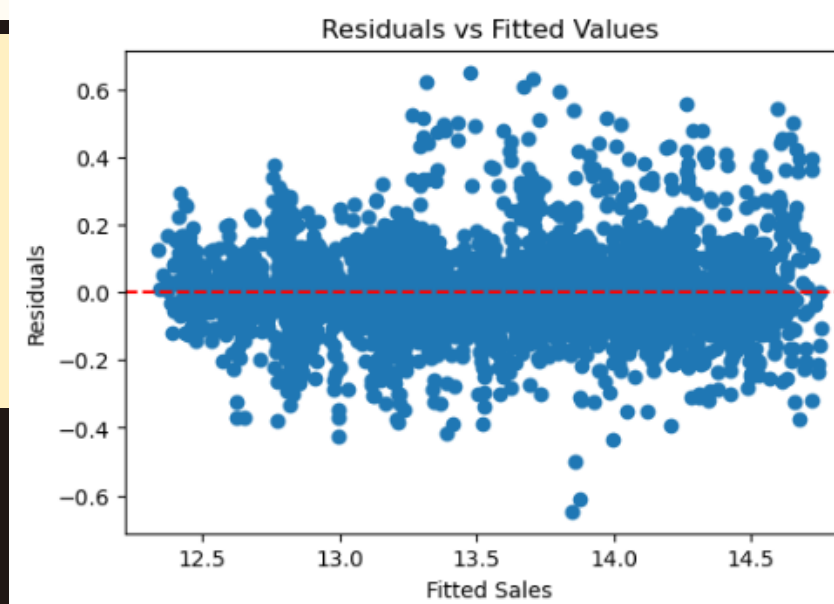


Validation

Linear ✓

Homoscedastous ✓

Multicollinearity ✗



Validation

Linear ✓

Homoscedastous ✓

Multicollinearity ✓

MAIN REVENUE DRIVERS

	coef	std err	t	P> t	[0.025	0.975]
const	14.3302	0.034	416.952	0.000	14.263	14.398
Holiday_Flag	0.0261	0.006	4.282	0.000	0.014	0.038
Temperature	0.0013	0.000	4.729	0.000	0.001	0.002
Fuel_Price	-0.0138	0.004	-3.300	0.001	-0.022	-0.006
Unemployment	-0.0274	0.003	-9.846	0.000	-0.033	-0.022
Day_Status_Month_Middle	0.0029	0.003	0.843	0.399	-0.004	0.010
Day_Status_Month_Start	0.0274	0.003	7.966	0.000	0.021	0.034

Train MAPE: 7.21%

Test MAPE: 7.59%

Training R2: 0.968

Testing R2: 0.968

Fuel Price Sensitivity: A \$1.00 increase in gas prices creates a -1.4% drag on sales

Paycheck Effect : Sales during Days 1 to 10 are 2.74% higher than the rest of the month

Holiday Boost: Being a "Holiday Week" yields a 2.6% lift, regardless of other factors.

Recession Drag: A 1% rise in Unemployment drops sales by 2.74%.

Recommendations

Shift restocking schedules to arrive by previous month-end. Ensure shelves are fully stocked for days 1 → 10 to prevent stockouts when customers have cash.

Integrate county-level unemployment data into workforce planning. If local unemployment rises, automatically reduce shift hours to protect margins



Since Fuel prices are very significant to the customer base, a tieup could be done with fuel pumps, so that customers (with Walmart membership) could get a slight % discount on the day they have purchased

Provide offers to people who buy in masse during the festive seasons (like christmas or other regional festivals (as holidays would be more during this time))



THANK YOU

ROSHAN PRAKASH